SEMESTER –VII

Sl No.	Course Code	Course Title	Hours Per Week			Total Credits
			Lecture	Tutorial	Practical	
1.	100701	Biology for Engineers	3	0	0	3
2.	153701	Data Analytics Using R Programming	3	0	0	3
3.	1537XX	Open Elective-I	3	0	0	3
4.	1537XX	Program Elective-IV	3	0	0	3
5.	153701P	Data Analytics Using R Programming Lab	0	0	2	1
6.	100702P	Summer Entrepreneurship - III	-	-	-	8
7.	100703P	Project-I	-	-	-	6
		TOTAL				27

List of Open Elective Courses - I

sr.no.	Code	Course Title	L	T	P	Credit
1.	153702	Embedded Systems and Hardware Security	3	0	0	3
2.	153703	Intellectual Property Rights	3	0	0	3
3.	153704	Virtual Reality	3	0	0	3
4.	153705	Electronic Circuits	3	0	0	3
5.	153706	Microprocessor	3	0	0	3

List of Program Elective Courses - IV

Sr. No.	Code	Course Title	L	T	P	Credit
1.	153707	Internet of Things	3	0	0	3
2.	153708	Optimization Techniques	3	0	0	3
3.	153709	High Performance Computing	3	0	0	3
4.	153710	Malware Analysis	3	0	0	3
5.	153711	Operation Research	3	0	0	3

SEMESTER – VII

Course Code- 100701 Biology for Engineers 3 0 0 3

Unit 1.0: Introduction 5 hrs

Purpose: To convey that Biology is as important a scientific discipline as Mathematics, Physics and Chemistry Bring out the fundamental differences between science and engineering by drawing a comparison between eye and camera, Bird flying and aircraft. Mention the most exciting aspect of biology as an independent scientific discipline. Why we need to study biology? Discuss how biological observations of 18th Century that lead to major discoveries. Examples from Brownian motion and the origin of thermodynamics by referring to the original observation of Robert Brown and Julius Mayor. These examples will highlight the fundamental importance of observations in any scientific inquiry.

Classification

Purpose: To convey that classification per se is not what biology is all about. The underlying criterion, such as morphological, biochemical or ecological be highlighted. Hierarchy of life forms at phenomenological level. A common thread weaves this hierarchy Classification. Discuss classification based on (a) cellularity- Unicellular or multicellular (b) ultrastructure- prokaryotes or eucaryotes. (c) energy and Carbon utilization -Autotrophs, heterotrophs, lithotropes (d) Ammonia excretion – aminotelic, uricoteliec, ureotelic (e) Habitata- acquatic or terrestrial (e) Molecular taxonomy- three major kingdoms of life. A given organism can come under different category based on classification. Model organisms for the study of biology come from different groups. E.coli, S. cerevisiae, D. Melanogaster, C. elegance, A. Thaliana, M. musculus.

Unit 2.0: Genetics 7 hrs

Purpose: To convey that "Genetics is to biology what Newton's laws are to Physical Sciences" Mendel's laws, Concept of segregation and independent assortment. Concept of allele. Gene mapping, Gene interaction, Epistasis. Meiosis and Mitosis be taught as a part of genetics. Emphasis to be give not to the mechanics of cell division nor the phases but how genetic material passes from parent to offspring. Concepts Page 11 of 75 of recessiveness and dominance. Concept of mapping of phenotype to genes. Discuss about the single gene disorders in humans. Discuss the concept of complementation using human genetics.

Biomolecules

Purpose: To convey that all forms of life has the same building blocks and yet the manifestations are as diverse as one can imagine Molecules of life. In this context discuss monomeric units and polymeric structures. Discuss about sugars, starch and cellulose. Amino acids and proteins. Nucleotides and DNA/RNA. Two carbon units and lipids.

Unit 3.0: Enzymes 8 hrs

Purpose: To convey that without catalysis life would not have existed on earth Enzymology: How to monitor enzyme catalyzed reactions. How does an enzyme catalyze reactions. Enzyme classification. Mechanism of enzyme action. Discuss at least two examples. Enzyme kinetics and kinetic parameters. Why should we know these parameters to understand biology? RNA catalysis.

Information Transfer

Purpose: The molecular basis of coding and decoding genetic information is universal Molecular basis of information transfer. DNA as a genetic material. Hierarchy of DNA structure- from single stranded to double helix to nucleosomes.

Concept of genetic code. Universality and degeneracy of genetic code. Define gene in terms of complementation and recombination.

Unit 4.0: Macromolecular analysis 7 hrs

Purpose: How to analyses biological processes at the reductionistic level

Proteins- structure and function. Hierarch in protein structure. Primary secondary, tertiary and quaternary structure. Proteins as enzymes, transporters, receptors and structural elements.

Unit 5.0: Metabolism 7 hrs

Purpose: The fundamental principles of energy transactions are the same in physical and biological world.

Thermodynamics as applied to biological systems. Exothermic and endothermic versus endergonic and exergoinc reactions. Concept of Keq and its relation to standard free energy. Spontaneity. ATP as an energy currency. This should include the breakdown of glucose to CO_2 + H_2O (Glycolysis and Krebs cycle) and synthesis of glucose from CO_2 and H_2O (Photosynthesis). Energy yielding and energy consuming reactions. Concept of Energy charge

Unit 6.0: Microbiology 8 hrs

Concept of single celled organisms. Concept of species and strains. Identification and classification of microorganisms. Microscopy. Ecological aspects of single celled organisms. Sterilization and media compositions. Growth kinetics.

Plant Physiology covering, Transpiration; Mineral nutrition. Ecology covering, Ecosystems- Components, types, flow of matter and energy in an ecosystem; Community ecology Characteristics, frequency, life forms, and biological spectrum; Ecosystem structure-Biotic and a-biotic factors, food chain, food web, ecological pyramids.

Text/ References Book:

1. Biology: A global approach: Campbell, N. A.; Reece, J. B.; Urry, Lisa; Cain, M, L.; Wasserman, S. A. Minorsky, P. V.; Jackson, R. B. Pearson Education Ltd.
2. Outlines of Biochemistry, Conn, E.E; Stumpf, P.K; Bruening, G; Doi, R.H., John Wiley and Sons.
3. Principles of Biochemistry (V Edition), By Nelson, D. L.; and Cox, M. M.W.H. Freeman and Company.
4. Molecular Genetics (Second edition), Stent, G. S.; and Calender, R.W.H. Freeman and company Distributed by Satish Kumar Jain for CBS Publisher.
5. Microbiology, Prescott, L.M J.P. Harley and C.A. Klein 1995. 2nd edition Wm, C. Brown Publishers.

Course Code- 153701 Data Analytics Using R Programming 3 0 0 3

Unit 1.0: Introduction to R 7 hrs

What is R? – Why R? – Advantages of R over Other Programming Languages - R Studio: R command Prompt, R script file, comments – Handling Packages in R: Installing a R Package, Few commands to get started: installed. packages(), package Description(), help(), find. package(), library() - Input and Output – Entering Data from keyboard – Printing fewer digits or more digits – Special Values functions : NA, Inf and –inf.

Unit 2.0: Data Types 7 hrs

Vectors, Lists, Matrices, Arrays, Factors, Data Frame – R - Variables: Variable assignment, Data types of Variable, Finding Variable Is(), Deleting Variables - R Operators: Arithmetic Operators, Relational Operators, Logical Operator, Assignment Operators, Miscellaneous Operators – R Decision Making: if statement, if – else statement, if – else if statement, switch statement – R Loops: repeat loop, while loop, for loop - Loop control statement: break statement, next statement.

Unit 3.0: R-Function 7 hrs

Function definition, Built in functions: mean(), paste(), sum(), min(), max(), seq(), user -defined function, calling a function, calling a function without an argument, calling a function with argument values - R-Strings – Manipulating Text in Data: substr(), strsplit(), paste(), grep(), toupper(), tolower().

Unit 4.0: R Vectors and Lists 6 hrs

R Vectors – Sequence vector, rep function, vector access, vector names, vector math, vector recycling, vector element sorting - R List - Creating a List, List Tags and Values, Add/Delete Element to or from a List, Size of List, Merging Lists, Converting List to Vector - R Matrices – Accessing Elements of a Matrix, Matrix Computations: Addition, subtraction, Multiplication and Division- R Arrays: Naming Columns and Rows, Accessing Array Elements, Manipulating Array Elements, Calculation Across Array Elements - R Factors –creating factors, generating factor levels gl().

Unit 5.0: Data Frames 8 hrs

Create Data Frame, Data Frame Access, Understanding Data in Data Frames: dim(), nrow(), ncol(), str(), Summary(), names(), head(), tail(), edit() functions - Extract Data from Data Frame, Expand Data Frame: Add Column, Add Row - Joining columns and rows in a Data frame rbind() and cbind() – Merging Data frames merge() – Melting and Casting data melt(), cast(). Loading and handling Data in R: Getting and Setting the Working Directory – getwd(), setwd(), dir() - R-CSV Files - Input as a CSV file, Reading a CSV File, Analyzing the CSV File: summary(), min(), max(), range(), mean(), median(), apply() - Writing into a CSV File R -Excel File – Reading the Excel file.

Unit 6.0: Descriptive Statistics 7 hrs

Data Range, Frequencies, Mode, Mean and Median: Mean Applying Trim Option, Applying NA Option, Median - Mode - Standard Deviation – Correlation – Spotting Problems in Data with Visualization: visually Checking Distributions for a single Variable - R –Pie Charts: Pie Chart title and Colors – Slice Percentages and Chart Legend, 3D Pie Chart – R Histograms – Density Plot - R – Bar Charts: Bar Chart Labels, Title and Colors.

Text/Reference Books

1. SandipRakshit, R Programming for Beginners, McGraw Hill Education (India), 2017, ISBN : 978-93-5260-455-5.
2. SeemaAcharya, Data Analytics using R, McGrawHill Education (India), 2018, ISBN: 978-93-5260-524-8.
3. Hadley Wickham and Garrett Grolemund, R for Data Science, O"Reilly Media, Inc. 2016, ISBN : 9781491910344.
4. Dr. Jeeva Jose, Beginner's Guide for Data Analysis using R Programming, Khanna Book Publishing, 2018, ISBN : 978-9386173454.
5. Andrie de Vries, JorisMeys, R for Dummies A Wiley Brand, 2nd Edition, John Wiley and Sons, Inc, 2015, ISBN: 978-1-119-05580-8.

Course Code- 153703 Intellectual Property Rights 3 0 0 3

Unit 1.0 7 hrs

Introduction to Intellectual property, need and importance intellectual property right (IPR), types of intellectual property: Patents, Trademarks, Copyrights, Industrial Designs, Geographical Indication, Semiconductor Circuits & Layout Design, Plant Variety & Farmer's Right Act, Genetic Resource and Traditional Knowledge, Undisclosed Information - Trade Secrets, IPR in abroad, International organizations, agencies and treaties, Some important examples of IPR and E-Commerce, IPR in Information Centric Environment, Career in IPR sector.

Unit 2.0 7 hrs

Concept of Copyright - Subject matter of copyright: original literary, dramatic, musical, artistic works; cinematograph films and sound recordings, Copyright Registration Procedure, Term of protection, Ownership of copyright, Assignment and licence of copyright - Infringement, Remedies & Penalties - Related Rights - Distinction between related rights and copyrights.

Unit 3.0 7 hrs

Concept of Trademarks, Different kinds of marks (brand names, logos, signatures, symbols, well known marks, certification marks and service marks), Non Registrable Trademarks, Procedure for Registration: Submission of application, acceptance, Advertisement, Rejection-Effect of Registration-Conditions for Registration-Registrar of Trademark. Rights of holder and assignment and licensing of marks.

Unit 4.0 6 hrs

Conccpt and History of Palcnts, Palcnt acts: India and abroad, Organisation of Patent systems in India, Indian Patents Act 1970 and subsequent amendments, WTO-TRIPS - Key effect on Indian Legislation, Rights and obligations of a patentee, Infringement, appellate board.

Unit 5.0 9 hrs

Patent Search: Overview of Patent Search , Advantages of patent search, Open source and paid databases for Patent Search, Patent Search Methodology, International Patent classification system. Drafting: Pre drafting requirement, Types of specifications, Drafting of Provisional specifications, Drafting of complete specifications, Drafting of claims. Filing procedure in India: Procedures for Patenting-Process of Obtaining a Patent-Application-Examination-Acceplance Opposition-Sealing of Patents-Preservation of Patents-Documentation-Register of Patents. PCT International Phase application, PCT National Phase application, Patent of addition, Divisional application.

Unit 6.0 6 hrs

Industrial Design: meaning and concept of novel and original -Procedure for registration, effect of registration and terms of protection. Integrated Circuit designs (Layout 1C Circuits) protection: meaning- Procedure for registration, effect of registration and term of protection. Case studies in IPR, Layout Design Appellate Board.

Text/Reference Books

1. Narayanan, P. (2010). Law of Copyright and Industrial Designs; Eastern law House, Delhi
2. Wadehra, B.L. (2000) Law Relating to Patents, Trade Marks, Copyright, Designs & Geographical Indications; Universal law Publishing Pvt. Ltd., India,

3. Nithyananda, K V. (2019). Intellectual Property Rights: Protection and Management, Cengage Learning India Private Limited.

Program Elective-IV

Course Code- 153707 **Internet of Things** **3 0 0 3**

Unit 1.0 **7 hrs**
Definition and Characteristics of IoT, Physical Design of IoT, Things in IoT, IoT Protocols, Logical Design of IoT: IoT Functional Blocks, IoT Communication Models and APIs, IoT Enabling Technologies, IoT Levels and Deployment Templates.

Unit 2.0 **6 hrs**
Sensors & actuators: Sensor working, sensor characteristics, types of sensors and working principle, sensors used in IoT.

Unit 3.0 **8 hrs**
Domain Specific IoT: Home Automation, Logistics, Agriculture, Introduction to M2M, Difference between IoT to M2M, IoT Design Methodology, Need for IoT System Management, IoT design methodology, case Study on IoT System for Weather Monitoring.

Unit 4.0 **6 hrs**
Introduction to Python, Python Data Types, Python Data Structures, Control Flow, Functions, Modules, Packages, File Handling, Classes, Python Database Programming, Python Packages for IoT.

Unit 5.0 **9 hrs**
IoT Physical Devices; Basic Building Blocks of an IoT device, Exemplary Device Raspberry Pi, Arduino Board, Beaglebone Black, Cubieboard.

Unit 6.0 **6 hrs**
Django- Python Web Application Framework, Designing RESTful Web API, Designing a Prototype for Smart City, Smart Parking and Smart Irrigation.

Text/Reference Books

1. Vijay Madisetti and Arshdeep Bahga, "Internet of Things (A Hands-on Approach)", 1st Edition, VPT, 2014.
2. Robert Barton, Patrick Grossetete, David Hanes, Jerome Henry, Gonzalo Salgueiro, "IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things", First Edition, Cisco Press, USA.
3. Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence, 1st Edition, Academic Press, 2014.
4. Bernd Scholz-Reiter, Florian Michahelles, "Architecting the Internet of Things", ISBN 978-3-642-19156-5 e-ISBN 978-3-642-19157-2, Springer.
5. Daniel Minoli, "Building the Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications", ISBN: 978-1-118- 47347-4, Willy Publications.

Course Code- 153701P Data Analytics Using R Programming Lab 0 0 2 1

List of experiments (Any 10)

1. Write a program to check whether a year (integer) entered by the user is a leap year or not?
2. Write an R program to find the sum of natural without formula using the if-else statement and while loop.
3. Write a program that prints the grades of the students according to the marks obtained. The grading of the marks should be as follows. Marks Grades 800-1000 A+, 700 – 800 A, 500 – 700 B+, 400-500 B, 150 – 400 C, Less than 150 D.
4. Write an R program to make a simple calculator that can add, subtract, multiply and divide using switch cases and functions.
5. Write a program to perform searching within a list (1 to 50). If the number is found in the list, print that the search is successful otherwise print that the number is not in the list.
6. Create a list and data frame that stores the marks of any three subjects for 10 students. Find out the total marks, average, maximum marks and minimum marks of every subject.
7. Write the steps to import data from Excel to CSV files and apply data viewer functions like rm(), dim(), head(), tail(), sorting, filtering, searching to view few set of rows.
8. Write a program to create two 3 X 3 matrices A and B and perform the following operations:
   a. Transpose of the matrix.
   b. Addition.
   c. Subtraction
9. Write an R program to create a list containing strings, numbers, vectors and logical values and do the following manipulations over the list:
   a. Access the first element in the list
   b. Give the names to the elements in the list
   c. Add element at some position in the list
   d. Remove the element
   e. Print the fourth element
   f. Update the third element
10. Use any dataset air quality which has Daily air quality measurements in New York, May to September 1973. Create a histogram by suing appropriate arguments for the following statements:
    a. Assigning names, using the air quality data set
    b. Change colours of the Histogram
    c. Remove Axis and Add Labels to Histogram
    d. Change Axis limits of a Histogram
    e. Create a Histogram with density and Add Density curve to the Histogram

11. Design a data frame in R for storing about 20 employee details. Create a CSV file named "input.csv" that defines all the required information about the employee such as id, name, salary, start_date, dept. Import into R and do the following analysis.
    a. Find the total number rows & columns
    b. Find the maximum salary
    c. Retrieve the details of the employee with maximum salary
    d. Retrieve all the employees working in the IT Department
    e. Retrieve the employees in the IT Department whose salary is greater than 20000 and write these details into another file "output.csv".
12. Create a dataset or table ["Smart Phone"] in an excel sheet that stores the mobile information [price, company name, model, Sale Percentage] of five different companies. Store at least 20 rows. Write the scripts and find out the output for the following information.
    a. Maximum price of the mobile of each company
    b. Minimum price of mobile of each company
    c. Average price of mobile of each company
    d. Total Price of mobile of each company

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